IS-RAG: Image-Schematic Retrieval-Augmented Generation for Cognitive Search in Political Discourse

Author One[1] Author Two[1] Author Three[2]

[1] Institution One [2] Institution Two

{author1,author2}@inst1.edu author3@inst2.edu

Abstract

We introduce **IS-RAG**, a technique that augments standard dense retrieval with *image-schema cognitive boosting* grounded in the cognitive linguistics of Lakoff and Johnson (1980) and Talmy (1988). During indexing, each document chunk is annotated by an LLM-based *Cognitive Parser* that identifies image schemas (FORCE, CONTAINER, PATH) and their subtypes, storing them as structured metadata alongside two independent vector embeddings—one over raw text, one over the serialised cognitive structure. At query time, IS-RAG detects schemas in the query and applies a continuous proportional score boost to chunks that share the same cognitive structure, enabling *cross-domain* retrieval driven by abstract spatial metaphor rather than lexical overlap. We evaluate IS-RAG on a corpus of 121 Brazilian parliamentary speeches in **Portuguese** using TREC-style pooling and NDCG@5. Over 9 valid evaluation queries (3 FORCE + 2 CONTAINER + 3 PATH + 1 cross-schema), IS-RAG achieves a mean NDCG@5 of 0.7073 vs. 0.6947 for a dense-retrieval baseline ($\Delta = +0.013$). Schema-family-weighted analysis reveals that gains concentrate in FORCE queries ($\Delta = +0.210$) while CONTAINER and PATH show neutral or negative results ($\Delta_{\text{family-weighted}} = -0.017$). We document methodological findings including the requirement for *assertive metaphorical query formulation* and the non-annotability of conventionalised schemas.

1. Introduction

Retrieval-Augmented Generation (Lewis et al., 2020) typically grounds document retrieval in surface-level lexical or dense semantic similarity. This works well when query vocabulary overlaps with document vocabulary, but breaks down under a *lexical-cognitive gap*: the user and the document express the same conceptual structure using different words, often because both rely on underlying bodily metaphors that map physical spatial experience onto abstract domains (Johnson, 1987; Lakoff and Johnson, 1980).

Consider a query about *economic pressure on households* and a parliamentary speech about *institutions blocking each other in a constitutional dispute*. Both invoke the image schema FORCE/COMPULSION—an external agent exerting force on a patient—yet their vocabularies share little overlap, making standard dense retrieval unable to bridge the gap.

Image schemas (Feldman, 2006; Johnson, 1987) are pre-linguistic, embodied patterns (CONTAINER, PATH, FORCE) that recur as structural metaphors across every domain of abstract reasoning. We hypothesise that indexing these schemas explicitly and matching them at query time enables a form of deep structural retrieval that complements dense semantic search.

Contributions.

- We present IS-RAG, a technique that indexes image-schema metadata alongside dual vector embeddings (text and cognitive) and applies proportional cognitive boosting at retrieval time (§3).
- We construct a TREC-style ground truth over a real corpus of Brazilian parliamentary speeches (171 chunks, 9 valid queries, LLM-as-annotator) and report NDCG@5 (§4).
- We identify methodological constraints for schema-aware retrieval evaluation: assertive query formulation, schema frequency vs. annotability, and LLM circularity risk (§6).

2. Related Work

Retrieval-Augmented Generation. Lewis et al. (2020) introduced RAG to ground generation in retrieved passages. Dense retrieval over bi-encoder representations (Karpukhin et al., 2020) has become the standard backbone. Extensions explore hybrid sparse-dense search, re-ranking, and structured knowledge augmentation, but none incorporate pre-linguistic spatial metaphor as a retrieval signal.

Metaphor in NLP. Automatic metaphor detection has been studied as classification or span extraction (Shutova, 2010; Tsvetkov et al., 2014). Wachowiak and Gromann (2022, 2023) explore LLMs for metaphor identification. These works focus on metaphor *detection*; IS-RAG is the first, to our knowledge, to use detected image-schema structure as a *retrieval scoring signal*.

Cognitive Linguistics and IR. Image schemas as cognitive universals were formalised by Lakoff and Johnson (1980), Johnson (1987), and Talmy (1988). Feldman (2006) discusses their neural basis. No prior IR work indexes documents by image schema and uses schema matching for retrieval boosting.

Portuguese NLP and parliamentary corpora. Pre-trained language models for Brazilian Portuguese (Souza et al., 2020) have demonstrated that language-specific pretraining substantially improves downstream performance relative to generic multilingual models. Parliamentary discourse in Portuguese has been studied through the Corpus Ulysses (Albuquerque et al., 2022), a collection of legislative documents from the Brazilian Chamber of Deputies used for text classification, named entity recognition, and semantic analysis. Our evaluation corpus is drawn from the same institutional source via the same open API and extends existing resources with image-schema annotations and graduated relevance judgements for retrieval evaluation. Framing and sentiment analysis of Brazilian political text (Vargas et al., 2021) provide further precedent for computational approaches to ideological structure in Portuguese parliamentary language.

3. Method

Figure 1 illustrates the IS-RAG pipeline. Formal pseudo-code is given in Appendix B.

[Figure — IS-RAG architecture: **left** = indexing pipeline (corpus → chunking → dual embeddings + cognitive parser → PostgreSQL/pgvector); **right** = query-time retrieval (query → cognitive analysis → hybrid scoring + boosting → ranked results).]

Figure 1: IS-RAG overview: indexing and retrieval steps.

3.1. Corpus and Chunking

The corpus comprises 121 Brazilian parliamentary speeches in **Portuguese**, collected from the open API of the Brazilian Chamber of Deputies (`dados.camara.leg.br/api/v2`), spanning January–December 2025. Speeches are stratified by ideological spectrum (left, centre, right) across 13 parties. Each speech is segmented into chunks of approximately 150 words using sentence-boundary splitting, yielding 171 chunks (mean 2.7 per speech).

3.2. Dual Embeddings

Each chunk produces two independent 768-dimensional vectors:

1. **Text embedding:** dense representation of the raw chunk text via *paraphrase-multilingual-mpnet-base-v2* (Reimers and Gurevych, 2019), running locally on CPU.
2. **Cognitive embedding:** dense representation of the serialised cognitive metadata in the form `"SCHEMA: anchor → domain | ..."`, capturing image-schematic structure independently of literal vocabulary.

Both vectors are stored in PostgreSQL 16 with the `pgvector` extension and indexed via HNSW for approximate nearest-neighbour search.

We deliberately chose `paraphrase-multilingual-mpnet-base-v2` over more recent commercial embedding APIs for two reasons. First, *reproducibility*: all embeddings are generated locally without API access, rate limits, or usage costs, enabling full replication of the experiment. Second, *adequacy*: the model provides validated coverage of Portuguese within a 768-dimensional space sufficient for the proof-of-concept scale of this work (171 chunks), and its multilingual training includes Brazilian Portuguese (Souza et al., 2020). We do not claim that upgrading to larger or more recent embedding models would not improve results; this is an explicit agenda item (§7) but is orthogonal to the variable under investigation—the cognitive boosting signal itself.

3.3. Cognitive Parser

The Cognitive Parser sends each chunk to `claude-haiku-4-5` (Anthropic, 2024) using a shared prompt (Appendix A) and extracts structured JSON validated by a Pydantic schema. The taxonomy follows Lakoff and Johnson (1980) and Talmy (1988):

Schema	Subtypes
CONTAINER	Inside, Outside, Boundary, Intrusion
PATH	Source, Trajectory, Goal, Diversion
FORCE	Blockage, Compulsion, Resistance, Counter_Force

A *literality rule* instructs the model to return empty lists for non-metaphorical text, preventing hallucinated schemas. Chunks are processed concurrently (5 workers) to reduce ingestion time. The corpus yields 92 PATH, 88 CONTAINER, and 83 FORCE schema occurrences across the 171 chunks.

3.4. Hybrid Search and Cognitive Boosting

IS-RAG supports three retrieval modes: *text* (text embedding only), *cognitive* (cognitive embedding only), and *hybrid* (mean of both similarities). After computing the base vector similarity s, a *proportional cognitive boost* is applied:

$$\hat{s} = s \cdot \big(1 + 0.4\,\sigma + 0.3\,\delta\big) \tag{1}$$

where σ is the fraction of a chunk's cognitive details whose `schema_name` matches a schema detected in the query, and δ is the analogous fraction for `target_domain`. Weights (α=0.4, β=0.3) were set by design; schema structure is the primary signal, domain the secondary reinforcement. The baseline disables the Cognitive Parser and applies pure text-embedding retrieval without boosting.

4. Evaluation Setup

Ground Truth Construction. We construct a TREC-style pooled ground truth (Voorhees and Harman, 2005). For each query, IS-RAG (hybrid) and the baseline independently retrieve the top-10 chunks; the union ($\approx$14–18 unique chunks per query) is annotated for relevance (0–3) by an LLM (`claude-haiku-4-5`). Scores ≥ 2 require the annotator to cite an anchor word and schema subtype as traceable evidence. Queries with IDCG$=0$ (no chunk with score ≥ 2) are excluded (Buckley and Voorhees, 2004); this criterion triggered the exclusion of one query (see §6).

Query Design. Nine valid queries are designed with three properties: (1) taxonomic coverage (3 FORCE + 2 CONTAINER + 3 PATH + 1 cross-schema); (2) deliberate lexical gap (queries use assertive metaphorical language, avoiding the literal vocabulary expected in documents); (3) cross-domain scenarios (queries target schemas that appear in domains other than the query's own domain).

Metric. We report NDCG@5 (Järvelin and Kekäläinen, 2002). MRR was rejected: with multiple relevant documents at different relevance grades, MRR saturates at 1.0 regardless of ranking quality (Buckley and Voorhees, 2004).

5. Results

5.1. Per-Query Results

Table 1 reports NDCG@5 for IS-RAG and the baseline across all 9 valid queries (hybrid mode). Figure 2 visualises the per-query Δ.

Q	Schema / Domain	IS-RAG	Base	Δ
Q2	FORCE/Compulsion / Economics	0.8375	0.5308	**+0.307**
Q3	FORCE/Counter_F / Justice	0.9093	0.6590	**+0.250**
Q4	FORCE/Resistance / Politics	0.6548	0.5818	+0.073
Q5	CONT./Intrusion / Justice	0.6403	0.9105	−0.270
Q6	CONT./Outside / Human Rights	0.7833	0.7226	+0.061
Q7	PATH/Trajectory / Economics	0.6488	0.6488	±0.000
Q8	PATH/Source / Politics	0.5745	0.6972	−0.123
Q9	PATH/Diversion / Politics	0.7777	0.8527	−0.075
Q10	FORCE+PATH / Intl. Rel.	0.5397	0.6488	−0.109
Mean (9 queries)		**0.7073**	**0.6947**	**+0.013**

Table 1: NDCG@5 (IS-RAG vs. baseline, hybrid mode). Queries grouped by schema family.

[Figure — Bar chart: Δ NDCG@5 per query (Q2–Q10), coloured by schema family — FORCE (blue), CONTAINER (orange), PATH (green), Cross (purple). Horizontal reference line at $\Delta = 0$.]

Figure 2: Per-query Δ NDCG@5 (IS-RAG − baseline).

5.2. Per-Schema Analysis

Table 2 aggregates results by schema family. The unweighted mean (Δ=+0.013) masks strong heterogeneity: FORCE drives all the gains, while CONTAINER and PATH are neutral or negative. Equal-weighting across the four schema families yields Δ_{family}=−0.017.

Family	n	Mean Δ	Individual Δ
FORCE	3	**+0.210**	+0.307, +0.250, +0.073
CONTAINER	2	−0.105	−0.270, +0.061
PATH	3	−0.066	0.000, −0.123, −0.075
Cross	1	−0.109	−0.109
Family-weighted mean		−0.017	

Table 2: NDCG@5 Δ by schema family (hybrid mode).

6. Discussion

Force benefits most from cognitive boosting. All three FORCE queries show positive Δ, with COMPULSION (+0.307) and COUNTER_FORCE (+0.250) as the strongest results. FORCE schemas tend to use salient, non-conventionalised metaphorical vocabulary in parliamentary discourse (*pressiona, bloqueia, choca*), making schema detection and boosting effective. This pattern is corpus-specific and should not be generalised without further validation.

Path and Container do not benefit. PATH metaphors appear conventionalised in parliamentary register to the point that they do not discriminate relevant from non-relevant chunks. CONTAINER/INTRUSION (Q5) was designed for *cognitive* mode; forced execution in hybrid mode explains part of the degradation ($\Delta = -0.270$).

Assertive query formulation is required. The Cognitive Parser applies a literality rule: interrogative or analytical queries (*"Which deputies argue that... "*) are correctly classified as non-metaphorical, returning empty schema lists and zeroing the boost. Queries must be formulated as *assertive metaphorical statements* to activate the cognitive component. This is a binding operational constraint for any IS-RAG deployment.

Schema frequency $\neq$ annotability. Q1 was designed as CONTAINER/INSIDE, the most frequent subtype in the corpus (80 occurrences). Despite high frequency, the annotator consistently rated INSIDE chunks at score 1 (domain present, schema non-salient) rather than ≥ 2 (clear metaphorical instantiation). Spatial containment is so conventionalised in parliamentary discourse that it does not constitute a discriminative relevance signal. Q1 was excluded (IDCG=0), yielding a final 3-2-3-1 query distribution.

Limitations. (1) The corpus is small and homogeneous (171 chunks, single register, Portuguese only); results lack statistical significance tests. (2) *LLM circularity*: the same model annotates, parses, and boosts—creating risk that the annotator favours chunks the parser indexed with the same schemas as the query. A blind human audit on a 20% sample of the annotated pool will be conducted in future work to compute Fleiss' Kappa inter-annotator agreement between independent human judges and the LLM annotator. (3) Domain confounding: FORCE queries cluster in Economics and Justice; if the schema co-occurs with those domains in the corpus, gains may reflect domain affinity rather than purely cognitive structure. (4) Boosting weights (α=0.4, β=0.3) are fixed design choices, not optimised.

7. Conclusion

This paper presents IS-RAG as a pioneering pilot study opening a new line of research at the intersection of cognitive linguistics and information retrieval. The core contribution is not a claim of universal superiority, but a demonstration that image-schema metadata can be encoded, detected, and leveraged within a standard dense retrieval pipeline—producing measurable NDCG@5 gains for FORCE-schema queries in hybrid mode ($\Delta_{\text{FORCE}} = +0.210$) on a real Brazilian parliamentary corpus.

The asymmetric results across schema families are themselves a finding: they reveal that schema frequency in a corpus does not predict annotability, and that the relationship between cognitive structure and retrieval relevance is non-trivial and worth systematic investigation. IS-RAG establishes the proof of concept and the methodological scaffolding—dual embeddings, cognitive boosting, TREC-style schema-stratified evaluation—that future work in *cognitive IR* can build upon.

Immediate next steps include a schema×domain crossed evaluation design, investigation of conventionalised schemas in graded-relevance annotation, cross-register and cross-language validation, and the Fleiss' Kappa human audit described above. We invite the IR and cognitive linguistics communities to extend, challenge, and refine this technique across richer corpora and retrieval architectures.

Code and Data Availability. Code, corpus, and annotated ground truth are publicly available at https://github.com/yanjustino/is-rag.

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A. Cognitive Analysis Prompt

The following prompt is shared between the indexing pipeline (`ingestion.py`) and the query-analysis step (`search.py`). It is written in **Portuguese** to match the language of the corpus and to preserve terminological precision in the few-shot examples. The placeholder `{text}` is replaced at runtime with the chunk or query text.

```
Você é um linguista cognitivo especialista na Teoria da
Metáfora Conceptual de Lakoff e nas Dinâmicas de Força de
Talmy. Sua missão é analisar um texto em português extraído
de discursos parlamentares, identificar os Esquemas
Imagéticos dominantes e mapear como eles estruturam
conceitos abstratos.

Analise o texto seguindo rigorosamente esta taxonomia:

1. MACROESQUEMAS E SUBTIPOS VÁLIDOS:
- CONTAINER:
  * INSIDE    (Estar contido, protegido, preso, engessado)
  * OUTSIDE   (Estar fora, excluído, marginalizado)
  * BOUNDARY  (Fronteiras, limites, barreiras de contenção)
  * INTRUSION (Invasão, penetração forçada no recipiente)
- PATH:
  * SOURCE     (Ponto de partida, origem, base histórica)
  * TRAJECTORY (Movimento, progresso, passos, rumo)
  * GOAL       (Destino, objetivo final, chegada)
  * DIVERSION  (Desvio de rota, perda de foco, sabotagem)
- FORCE:
  * BLOCKAGE      (Bloqueio completo, barreira física/legal)
  * COMPULSION    (Força externa que empurra, obriga, coage)
  * RESISTANCE    (Resistência interna, oposição ativa)
  * COUNTER_FORCE(Duas forças colidindo, embate direto)

2. REGRAS PARA target_domain_pt (lista fechada):
"Economia" | "Política" | "Infraestrutura e Transportes"
| "Segurança Pública" | "Justiça" | "Direitos Humanos e
Cultura" | "Educação" | "Saúde" | "Meio Ambiente"
| "Relações Internacionais" | "Outros"

3. REGRA DE LITERALIDADE:
Se o texto for puramente literal, descritivo, técnico ou
administrativo, retorne listas vazias.

EXEMPLOS (FEW-SHOT):
```

```
Texto: "A oposição barrou os avanços da reforma tributária."
Output:
{"schemas": ["FORCE","PATH"], "details": [
  {"schema_name":"FORCE","sub_type":"BLOCKAGE",
   "anchor_word_pt":"barrou",
   "target_domain_pt":"Política"},
  {"schema_name":"PATH","sub_type":"TRAJECTORY",
   "anchor_word_pt":"avanços",
   "target_domain_pt":"Economia"}
]}

Texto: "A inflação empurrou as famílias para a crise."
Output:
{"schemas": ["FORCE"], "details": [
  {"schema_name":"FORCE","sub_type":"COMPULSION",
   "anchor_word_pt":"empurrou",
   "target_domain_pt":"Economia"}
]}

Texto: "Quero saudar a presença do Vereador Júnior."
Output: {"schemas": [], "details": []}

---
Analise o texto a seguir e retorne o JSON estrito.

Texto para análise:
{text}
```

B. IS-RAG Algorithms

B.1. Indexing Pipeline

Algorithm 1 IS-RAG Indexing

Input: Corpus $\mathcal{D}$, chunk size w=150, encoder $\mathcal{E}$, LLM parser $\mathcal{P}$
Output: Populated vector store $\mathcal{S}$

for each document $d \in \mathcal{D}$ **do**
    $C \leftarrow \textsc{SemanticChunk}(d, w)$ ▷ sentence-boundary split
    $\mathbf{E}_t \leftarrow [\,\mathcal{E}(c) \mid c \in C\,]$ ▷ text embeddings, 768d
    $M \leftarrow [\,\mathcal{P}(c) \mid c \in C\,]$ ▷ concurrent LLM calls, 5 workers
    for $i \leftarrow 1$ **to** $|C|$ **do**
        $\mathbf{e}_c \leftarrow \mathcal{E}(\textsc{Serialize}(M_i))$ ▷ cognitive embedding
        $\mathcal{S}.\textsc{Insert}(C_i, \mathbf{E}_{t,i}, \mathbf{e}_c, M_i)$
    end for
end for

B.2. Query-Time Retrieval

Algorithm 2 IS-RAG Retrieval

Input: Query q, mode $m \in \{$text, cognitive, hybrid$\}$, top-k, store $\mathcal{S}$, α=0.4, β=0.3
Output: Ranked list of k chunks

$(\Sigma_q, \Delta_q) \leftarrow \mathcal{P}(q)$ ▷ detected schemas and domains
$\mathbf{e}_{q,t} \leftarrow \mathcal{E}(q);\ \mathbf{e}_{q,c} \leftarrow \mathcal{E}(\textsc{Serialize}(\mathcal{P}(q)))$
for each chunk $(C_i,\ \mathbf{e}_{t,i},\ \mathbf{e}_{c,i},\ M_i) \in \mathcal{S}$ **do**
  $s_t \leftarrow \cos(\mathbf{e}_{q,t},\ \mathbf{e}_{t,i});\ s_c \leftarrow \cos(\mathbf{e}_{q,c},\ \mathbf{e}_{c,i})$
  $s \leftarrow \begin{cases} s_t & \text{if } m = \text{text} \\ s_c & \text{if } m = \text{cognitive} \\ (s_t + s_c)/2 & \text{if } m = \text{hybrid} \end{cases}$
  $n \leftarrow |M_i.\text{details}|$
  $\sigma \leftarrow |\{d \in M_i.\text{details} \mid d.\text{schema} \in \Sigma_q\}|\ /\ \max(n, 1)$
  $\delta \leftarrow |\{d \in M_i.\text{details} \mid d.\text{domain} \in \Delta_q\}|\ /\ \max(n, 1)$
  $\hat{s}_i \leftarrow s \cdot (1 + \alpha\,\sigma + \beta\,\delta)$ ▷ Eq. 1
end for
return $\textsc{TopK}(\{\hat{s}_i\},\ k)$